

Knowledge Representation in AI

Logic: propositional, predicate, First Order Logic.

Normal forms. Modus Ponens & Modus Tollens,

Theorem Proving, Principle of Resolution, Non-Monotonic Reasoning

Semantic Net, Frame

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Types of knowledge

One of the major requirements in AI is the need to represent domain knowledge

1. Declarative/Descriptive Knowledge

Includes concepts, facts, and objects.

2. Procedural/Imperative Knowledge

Provides knowledge about how to do something.

Represented as ifthen ..else rules and procedures.

3. Heuristic knowledge

Heuristic knowledge is Rules of thumb based on previous experiences

4. Structural knowledge

Describes the relationship that exists between concepts or objects.

5. Meta-knowledge

Knowledge about other types of knowledge

Requirements of Knowledge Representation

1. **Acquisitional efficiency**

The ability to capture knowledge from the source using appropriate methods/tools.

2. **Representational Accuracy**

KR system should capture the required knowledge accurately.

3. **Inferential Adequacy**

KR system should have the ability to logically deduce new knowledge from the existing knowledge structure.

4. **Inferential Efficiency**

The ability to navigate the inferencing process in the most productive directions by using appropriate strategies.

Use of logic in Knowledge Representation

- Propositional logic (PL) is the simplest form of logic where all the statements are made by propositions. A proposition is a declarative statement which is either true or false.

P: It is sunny

Q: Sky is blue

- Several propositions can be connected by logical connectives (AND, NOT, OR)
- A proposition formula which is always true is called **tautology**.
- A proposition formula which is always false is called **Contradiction**.
- **Atomic proposition:** "It is raining today"
- **Compound proposition:** "It is raining and the road is wet."

Logical Connectives

- **Negation:** $\neg P$ is called negation of P.
- **Conjunction:** $P \wedge Q$
“Rohan is intelligent and hardworking”
P: Rohan is intelligent.
Q: Rohan is hardworking. $(P \wedge Q)$
- **Disjunction:** $P \vee Q$
"Ravi is a doctor or Engineer",
P: Ravi is a Doctor. Q: Ravi is an Engineer. $(P \vee Q)$
- **Implication:** $P \rightarrow Q$. Implications are also known as if-then rules.
If it is raining, then the street is wet.
If P: It is raining & Q: Street is wet then $P \rightarrow Q$
- **Biconditional:** $P \Leftrightarrow Q$,
If I am breathing, then I am alive (vice versa is also true)
P: I am breathing, Q: I am alive, then as $P \Leftrightarrow Q$

Limitations of Propositional logic

- Propositional logic has limited expressive power.
- We cannot represent relations like All, Some, or None with propositional logic.
 - **All children like chocolates.**
 - **Some students are intelligent.**
 - **None of the villagers have voted.**

Predicate logic(PL)/First-order predicate logic (FOPL) /First-order logic(FOL) - These terms are used interchangeably

It is expressive enough to represent natural language sentences in a precise manner.

A **predicate is a relation**, which binds two atoms together in a statement.

Rocky is a dog: $\Rightarrow$ dog (Rocky).

Amit & Arun are friends: $\Rightarrow$ friends(Amit, Arun)

- **Universal Quantifier: (for all, everyone, everything)**

$\forall x$ child (x) $\rightarrow$ likes (x, chocolate)

- **Existential quantifier: (for some, at least one)**

$\exists x$: student (x) $\wedge$ intelligent(x)

Note:

- The main connective for $\forall$ is implication $\rightarrow$.
- The main connective for $\exists$ is and $\wedge$.

Properties

- $\forall x \forall y$ is equivalent to $\forall y \forall x$.
- $\exists x \exists y$ is equivalent to $\exists y \exists x$.
- $\exists x \forall y$ is not equivalent to $\forall y \exists x$.

It is called first order because the quantifiers apply only to individual objects, and not over functions or sets of objects.

Examples of FOPL

1. All birds fly

Here, the predicate is "fly(bird)."

$$\forall x \text{ bird}(x) \rightarrow \text{fly}(x).$$

2. Every child respects his parent

Here, the predicate is "respect(x, y)," where x=child, and y= parent.

$$\forall x \text{ child}(x) \rightarrow \text{respects}(x, \text{parent}).$$

3. Some boys play cricket

Here, the predicate is "play(x, y)," where x= boys,& y= game.

$$\exists x \text{ boys}(x) \wedge \text{play}(x, \text{cricket}).$$

4. Not all students like both Mathematics and Science

Here, the predicate is "like(x, y)," where x= student, and y= subject.

$\neg \forall (x) [\text{student}(x) \rightarrow \text{like}(x, \text{Mathematics}) \wedge \text{like}(x, \text{Science})]$.

5. Only one student failed in Mathematics

Here, the predicate is "failed(x, y)," where x= student, and y= subject.

$\exists (x) [\text{student}(x) \rightarrow \text{failed}(x, \text{Mathematics})]$

Free and Bound Variables:

Free Variable: A variable is said to be a free variable in a formula if it occurs outside the scope of the quantifier.

Bound Variable: A variable is said to be a bound variable in a formula if it occurs within the scope of the quantifier.

$$\forall x \exists (y)[P(x, y, z)]$$

Here, z is a free variable

x & y are bound variables.

Inferencing in FOPL

The process of inferencing can lead to creation of new knowledge which is not known explicitly.

1. Universal Generalization:

It states that if premise $P(c)$ is true for any arbitrary element c in the universe of discourse, then we can conclude as $\forall x P(x)$.

$$\frac{P(c)}{\forall x P(x)}$$

Used to show that every element has a similar property.

Ex : $P(c)$: "A byte contains 8 bits",
so for $\forall x P(x)$ "All bytes contain 8 bits."

2. Universal Instantiation:

Also referred as *universal elimination*.

It states that we can infer any sentence $P(c)$ by substituting a ground term c (a constant within domain x) from $\forall x P(x)$ for any object in the universe of discourse.

$$\frac{\forall x P(x)}{P(c)}$$

Ex:1

If "Every person like ice-cream" $\Rightarrow \forall x P(x)$

we can infer that "John likes ice-cream" $\Rightarrow P(c)$

Ex: 2.

If "All kings who are greedy are Evil."

$$\forall x \text{ king}(x) \wedge \text{greedy}(x) \rightarrow \text{Evil}(x)$$

From this we can infer any of the following statements:

- $\text{King}(\text{Tom}) \wedge \text{Greedy}(\text{Tom}) \rightarrow \text{Evil}(\text{Tom})$
- $\text{King}(\text{Jerry}) \wedge \text{Greedy}(\text{Jerry}) \rightarrow \text{Evil}(\text{Jerry})$
- $\text{King}(\text{Father}(\text{Tom})) \wedge \text{Greedy}(\text{Father}(\text{Tom})) \rightarrow \text{Evil}(\text{Father}(\text{Tom}))$

3. Existential Instantiation:

It states that one can infer **P(c)** from the formula given in the form of **∃x P(x)** for a new constant symbol **c**.

$$\frac{\exists x P(x)}{P(c)}$$

From the given sentence: **∃x Crown(x) ∧ OnHead(x, John)**,

we can infer: **Crown(K) ∧ OnHead(K, John)**, as long as K does not appear in the knowledge base.

4. Existential introduction

It states that if there is some element **c** in the universe of discourse which has a property **P**, then we can infer that there exists something in the universe which has the property P.

$$\frac{P(c)}{\exists xP(x)}$$

"Ravi got good marks in Computer Science."

"Therefore, someone got good marks in Computer Science."